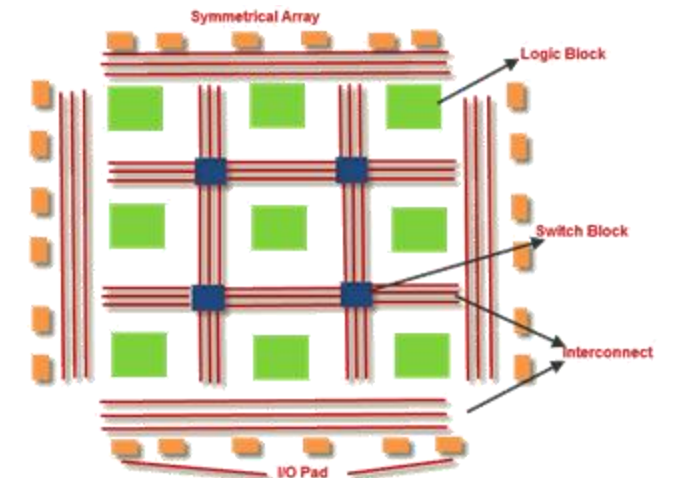
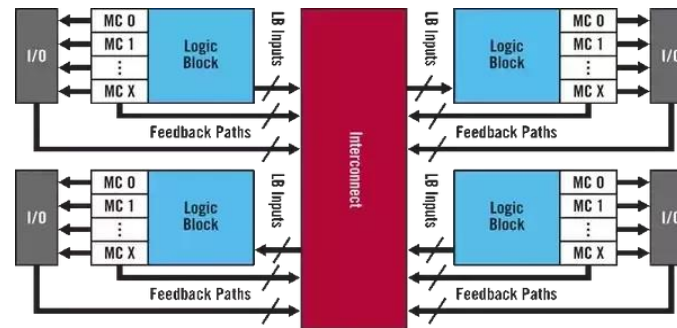
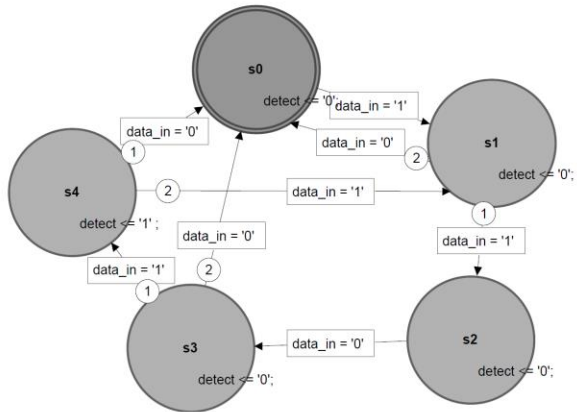
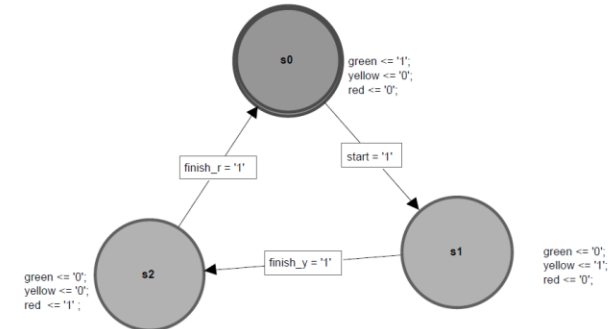
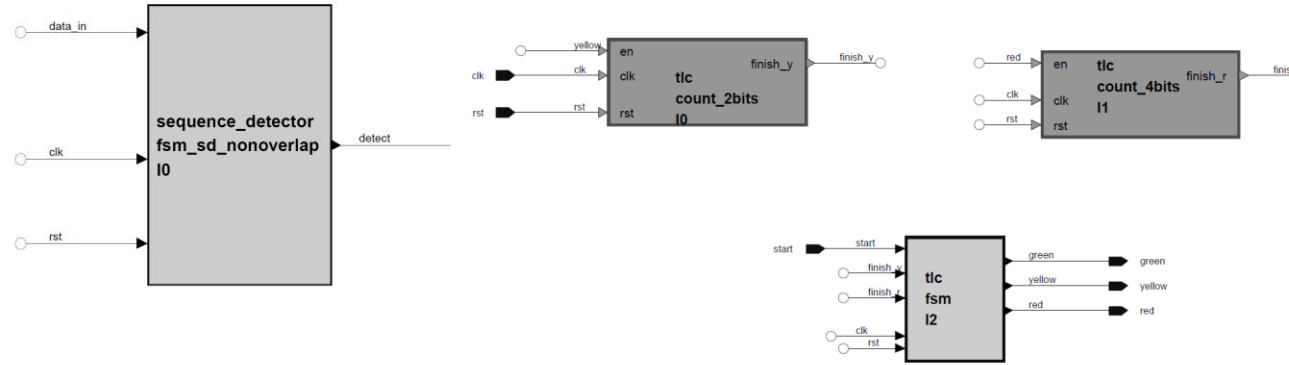


Lecture 8: FSM: keypad and Calculator

Vincent Helbig

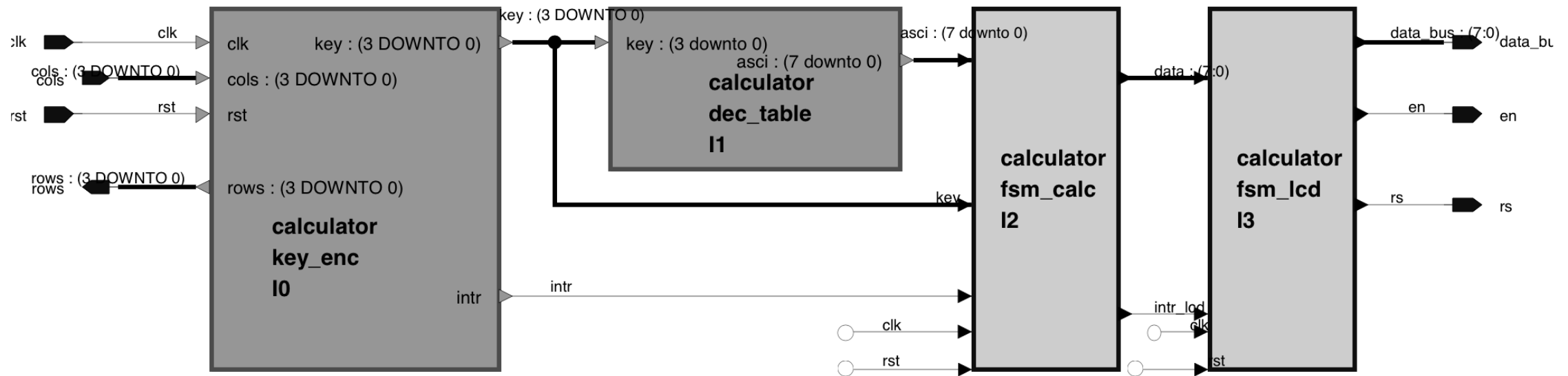
Original Author: Emad Samuel Malki Ebeid

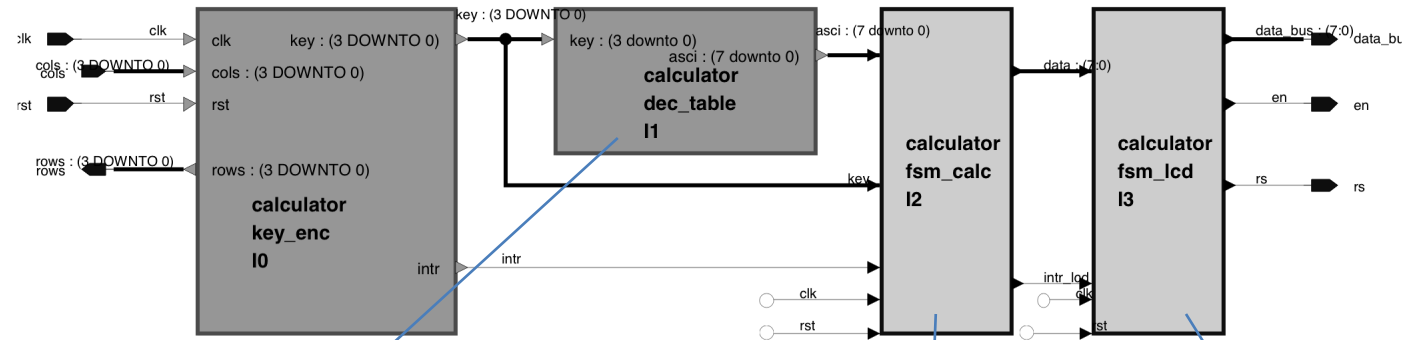
Summary of lecture 7



Simple Calculator with an LCD and a keypad

Simple Calculator with an LCD and a keypad



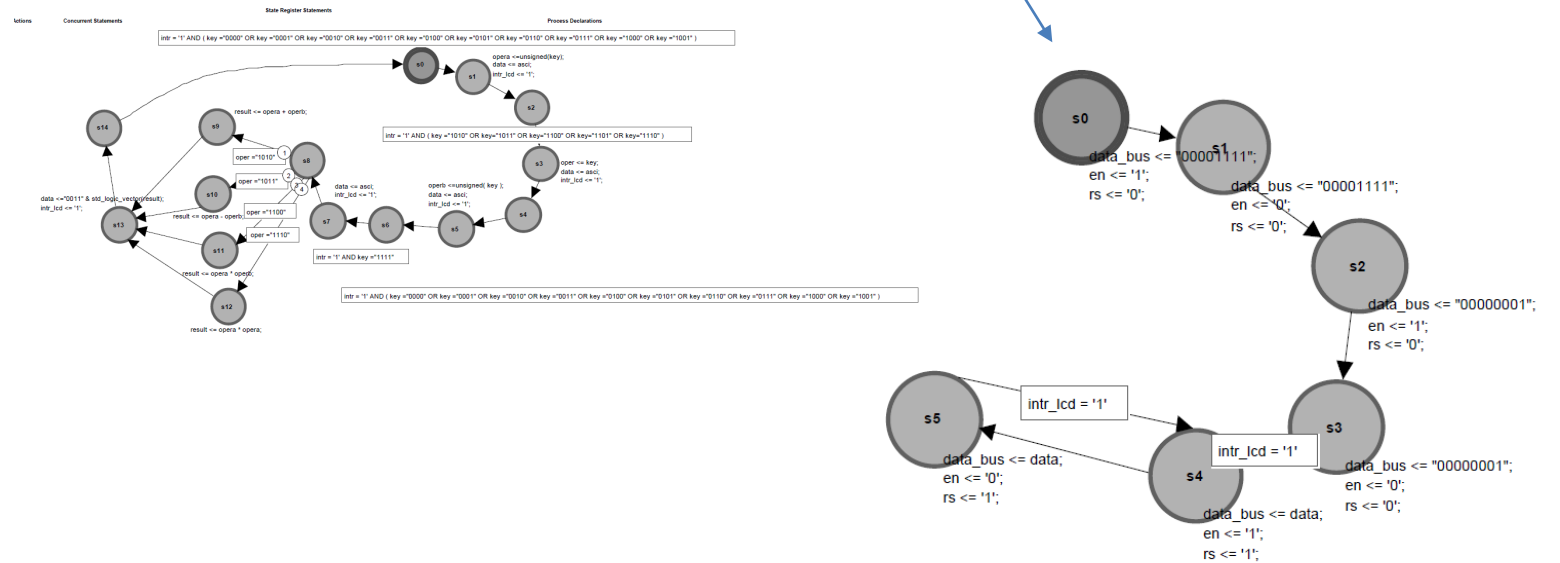


```

ENTITY dec_table IS
-- Declarations
  port(key: in std_logic_vector(3 downto 0);
        ascii : out std_logic_vector(7 downto 0));
END dec_table ;

-- hds interface_end
ARCHITECTURE rtl OF dec_table IS
BEGIN
  process(key)
  begin
    if (key="0000") then ascii<="00110000";
    elsif (key="0001") then ascii<="00110001"; --0
    elsif (key="0010") then ascii<="00110010";
    elsif (key="0011") then ascii<="00110011";
    elsif (key="0100") then ascii<="00110100";
    elsif (key="0101") then ascii<="00110101";
    elsif (key="0110") then ascii<="00110110";
    elsif (key="0111") then ascii<="00110111";
    elsif (key="1000") then ascii<="00111000";
    elsif (key="1001") then ascii<="00111001"; --9
    elsif (key="1010") then ascii<="00101011"; --+
    elsif (key="1011") then ascii<="00101101"; -- -
    elsif (key="1100") then ascii<="00101010"; -- *
    elsif (key="1101") then ascii<="00101111"; -- ^2
    elsif (key="1110") then ascii<="00101110"; --.
    else ascii<="00111101"; --=
    end if;
  end process;
END rtl;

```



FSM_calc

